Darshan Kapoor

Embedded Firmware Engineer

RESUME_1813

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Total Experience: 8 years

PROFESSIONAL SUMMARY

Embedded Firmware Engineer with 8 years of experience in embedded firmware, using C, C++, ARM Cortex-M, FreeRTOS. Experienced in real-time systems, hardware debugging and measurable delivery outcomes.

TECHNICAL SKILLS

Skills: ARM Cortex-M, Git, SPI, C, Oscilloscope, CAN, C++, I2C, UART, FreeRTOS, STM32, Firmware Debugging

Methods: Real-time systems, Hardware debugging, Protocol validation

PROFESSIONAL EXPERIENCE

Senior Embedded Firmware Engineer | Ironleaf Engineering | Jan 2023 - Present

- Reduced embedded task jitter by 38% using interrupt priority and queue tuning
- Lowered field-reset incidents by 31% with watchdog recovery and diagnostics
- Improved battery runtime by 25% with peripheral and sleep-state optimization
- Coordinated engineering design reviews, acceptance criteria and test evidence with cross-functional stakeholders.

Embedded Firmware Engineer | NovaPeak Engineering | Jan 2021 - Dec 2022

- Improved battery runtime by 16% with peripheral and sleep-state optimization
- Validated 96 hardware test vectors across protocol and timing corner cases

Associate Embedded Firmware Engineer | GlideArc Solutions | Jul 2018 - Dec 2020

- Reduced embedded task jitter by 39% using interrupt priority and queue tuning
- Lowered field-reset incidents by 23% with watchdog recovery and diagnostics

PROJECTS

Industrial Sensor Firmware | STM32, Firmware Debugging, FreeRTOS, I2C

- Built FreeRTOS tasks for calibrated sampling and watchdog recovery.
- Documented acceptance criteria and evaluated the design using hardware debugging and protocol validation.

Bootloader Update Utility | STM32, I2C, C++, Oscilloscope

- Developed verified firmware transfer and rollback checks.
- Documented acceptance criteria and evaluated the design using hardware debugging and protocol validation.

EDUCATION

B.Tech in Electronics and Communication Engineering, Bayview Institute of Engineering, 2018
Relevant Methods: Real-time systems, Hardware debugging, Protocol validation

CERTIFICATIONS

RTOS Scheduling Training (2026)